

Confidence-Gated Multi-Agent Fall Detection: Retrieval-Augmented Reasoning and Cost-Sensitive Response on Wearable IMU Data

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Abstract—Wearable inertial fall detectors achieve strong average accuracy but remain brittle on ambiguous near-fall activities and provide neither calibrated uncertainty routing nor an explicit response policy under asymmetric false-negative cost. We present a *confidence-gated multi-agent* architecture that wraps a fixed Tier-1 detector—instantiated as the CNN–LSTM–Attention backbone of Bhatti *et al.*—with (i) a dual-threshold confidence gate, (ii) retrieval-augmented k -nearest-neighbor case memory, (iii) local large-language-model (LLM) chain-of-thought reasoning over biomechanical evidence, and (iv) a cost-sensitive action agent. On SisFall under subject-independent five-fold cross-validation with a locked evaluation protocol, the full stack (`gate_knn_llm`) improves mean F1 from 0.759 ± 0.014 to 0.887 ± 0.029 and mean recall from 0.759 ± 0.022 to 0.977 ± 0.037 , while reducing normalized response cost by approximately 83% relative to the same Tier-1 checkpoint. Ablations show that gating or retrieval alone is insufficient; paired ambiguous-case comparisons further show that Mistral substantially outperforms a heuristic reasoner (mean F1 0.972 vs. 0.034). We position results as gains over a fair reimplementations of the Bhatti-style detector under our protocol, not as a claim against the original KFall multi-phase tables.

Index Terms—Fall detection, wearable IMU, confidence gating, retrieval-augmented reasoning, large language models, cost-sensitive decision making, SisFall.

I. INTRODUCTION

Population aging and independent living motivate continuous fall monitoring with wearable inertial measurement units (IMUs) [1]–[3]. Modern deep detectors can reach competitive accuracy on curated benchmarks, yet practical systems still fail in three ways that matter for safety-critical deployment. First, near-fall activities (e.g., sitting with compensatory motions) induce ambiguous windows that a softmax classifier force-labels. Second, edge models rarely expose calibrated uncertainty that can defer difficult cases. Third, a binary fall/ADL prediction is not a response policy: emergency escalation, caregiver notification, and logging have unequal costs, especially when false negatives dominate clinical risk.

Bhatti *et al.* [4] propose a strong hybrid CNN–LSTM–Attention detector for multi-phase fall analysis on wearable IMUs (“Beyond Falls”). Their contribution is primarily *detection*. We ask a complementary question: *given a fixed Tier-1 detector, can an agentic wrapper improve the safety operating point—raising recall and lowering expected response cost—while providing structured explanations on escalated windows?*

This paper answers affirmatively with a confidence-gated multi-agent pipeline. Uncertain predictions are escalated to an

evidence serializer, a growing k -NN memory of past cases, and a local LLM reasoner that emits JSON rationales; an Action Agent then maps decisions to simulated response tiers under cost $10 \cdot \text{FN} + 1 \cdot \text{FP}$. Our primary evaluation is SisFall [3] with subject-independent five-fold cross-validation, forced inclusion of near-fall codes D18/D19, and a homogeneous evaluation budget of 1000 stratified windows per fold.

a) Contributions.:

- A complete agentic architecture—gate, evidence, retrieval, LLM CoT, action—wrapped around a Bhatti-style Tier-1 backbone without modifying its weights at inference.
- A locked paper protocol (splits, calibration, cost, near-fall metrics) that enables fair ablations: `tier1_only` $\rightarrow$ `gate_only` $\rightarrow$ `gate_knn` $\rightarrow$ `gate_llm` $\rightarrow$ `gate_knn_llm`.
- Five-fold SisFall evidence that the full stack improves F1/recall and normalized cost on every fold, with paired LLM backend comparisons on shared ambiguous IDs.
- An honest positioning relative to Bhatti *et al.*: gains are versus the *same* Tier-1 checkpoint under our SisFall protocol; we do not claim to supersede the original KFall multi-phase tables.

II. RELATED WORK

A. Wearable Fall Detection

Classical threshold and shallow-learning detectors have been succeeded by CNN, LSTM, TCN, and Transformer variants on public corpora such as SisFall [3] and KFall [5]. Surveys emphasize the trade-off between sensitivity and false alarms in free living [1]. Bhatti *et al.* [4] integrate CNN, LSTM, and attention for pre-/transition-/post-fall phases. We reuse their architectural motif as Tier-1 and add decision-time agents rather than proposing another detector backbone as the sole contribution.

B. Uncertainty and Selective Prediction

Calibration and abstention [6] motivate routing low-confidence samples to stronger (or human) decision makers. Our dual-threshold gate implements selective escalation tailored to fall cost asymmetry.

C. Retrieval and LLM Agents for Sensing

Retrieval-augmented generation and compact encoders such as MiniLM [7] support case-based reasoning. Local LLM runtimes [8]–[10] enable offline CoT without cloud APIs. We apply these ideas to IMU evidence rather than text-native tasks.

III. METHOD

Fig. 1 summarizes the pipeline.

A. Tier-1 Detector

Given a window $\mathbf{x} \in \mathbb{R}^{C \times T}$ (here $C=6$, $T=90$), a backbone f_θ produces logits and embedding $\mathbf{e}$. We convert logits to fall probability $p(\text{fall})$. The primary backbone is CNN-LSTM-Attention following Bhatti *et al.* [4]; we additionally train a zoo (CNN1D, LSTM, CNN-LSTM, TCN, ModernTCN [11], TSMixer [12], PatchTST [13], Transformer [14]) under the same protocol for context (Table VI).

B. Confidence Gate

A dual-threshold gate with calibrated $(\tau_{\text{low}}, \tau_{\text{high}})$ routes windows:

$$\begin{cases} \text{ADL}, & p \leq \tau_{\text{low}}, \\ \text{Fall}, & p \geq \tau_{\text{high}}, \\ \text{Ambiguous}, & \text{otherwise.} \end{cases} \quad (1)$$

Thresholds are calibrated on subject-held-out training windows. Mode `tier1_only` disables meaningful escalation by collapsing the ambiguous band.

C. Evidence Serializer

For escalated windows we compute biomechanical descriptors (free-fall, impact magnitude, stillness, tilt) and serialize them into natural-language evidence for retrieval and prompting.

D. Retrieval Memory

A k -NN memory stores MiniLM [7] embeddings of evidence text with labels and features. At escalation time we retrieve $k=5$ neighbors (cosine similarity) as few-shot context.

E. LLM Reasoner

An LLM reasoner (`backend` $\in \{\text{ollama}, \text{heuristic}\}$) consumes evidence and neighbors and returns a JSON decision with rationale. We evaluate heuristic, Mistral [9], and Qwen2.5 [10] on paired ambiguous IDs. Production full-stack rows use Ollama [8].

F. Action Agent and Cost

The Action Agent maps severity cues to simulated actions (emergency / notify / monitor / log). Expected response cost uses

$$\text{Cost} = 10 \cdot \text{FN} + 1 \cdot \text{FP}. \quad (2)$$

We report *cost per 1000 windows* so folds with equal evaluation budgets are comparable.

G. System Modes

Ablations instantiate cumulative agents: `tier1_only`, `gate_only`, `gate_knn`, `gate_llm`, and `gate_knn_llm` (full stack).

IV. EXPERIMENTAL SETUP

A. Dataset and Splits

We use SisFall [3] windows with six IMU channels. Subject-independent five-fold splits hold out subjects; within each training fold a subject-held-out validation fraction tunes training and gate calibration. Near-fall activities D18/D19 are required and forced into capped test slices (minimum per code).

B. Protocol

Training uses class imbalance handling (`pos_weight`), mixup, and cost-sensitive decision thresholds. The **main paper tables** use a homogeneous evaluation budget of $n=1000$ stratified test windows per fold. An archived Fold 0 run with $n=4000$ provides a detailed ablation ladder including `gate_llm`. Seed is fixed (42).

C. Metrics

We report F1, recall/sensitivity, specificity, near-fall false-alarm rate on D18/D19, escalated-subset F1, escalation rate, and normalized response cost. Significance uses paired bootstrap 95% confidence intervals of mean fold-wise deltas; Wilcoxon signed-rank p -values are reported for completeness ($n=5$).

V. RESULTS

A. Main Comparison vs. Tier-1

Table II shows that `gate_knn_llm` improves F1 on every fold (+0.114 to +0.164). Table III summarizes five-fold means: F1 0.759→0.887, recall 0.759→0.977, cost/1000 1151→200 (about 83% reduction). Bootstrap CIs for ΔF1 , Δrecall , and $\Delta\text{cost}/1000$ exclude zero.

B. Ablations

Table IV (Fold 0, $n=4000$) shows that `gate_only` matches Tier-1 on aggregate F1, `gate_knn` does not unlock large gains, and `gate_llm` alone can inflate near-fall FAR. The full stack reaches F1 0.907, recall 0.993, and escalated F1 0.952 at substantially lower cost. Thus retrieval and LLM reasoning are complementary.

C. LLM Backends on Ambiguous Cases

Table V compares reasoners on shared ambiguous IDs. Heuristic mean F1 is near zero (0.034), while Mistral reaches 0.972 and Qwen2.5 0.823. This supports the claim that language-model reasoning—not merely escalation—drives ambiguous-case recovery.

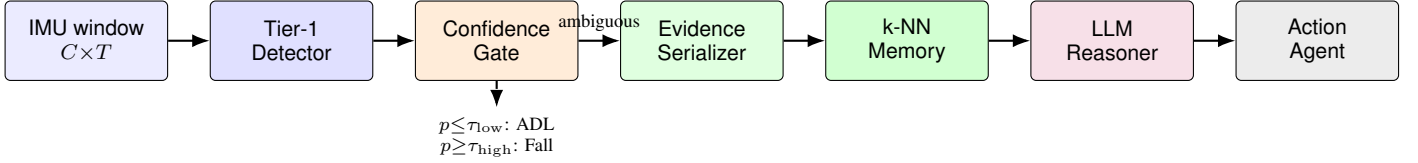


Fig. 1. Confidence-gated multi-agent fall detection. High/low confidence windows are decided by the Tier-1 detector; ambiguous windows escalate to evidence serialization, k -NN retrieval, LLM reasoning, and a cost-sensitive Action Agent.

TABLE I
POSITIONING RELATIVE TO BHATTI *et al.* (BASE DETECTOR PAPER).

Aspect	Bhatti <i>et al.</i> 2025 [4]	This work
Primary task	Multi-phase fall/ADL classification	Detection + uncertainty + explanation + response
Backbone	CNN–LSTM–Attention	Same primary backbone + modern zoo
Uncertainty	Softmax confidence only	Dual-threshold confidence gate (τ_{low}, τ_{high})
Explainability	Attention weights	Structured biomechanical evidence + CoT JSON
Case memory	None	Retrieval-augmented k -NN memory
Response policy	None	Cost-sensitive Action Agent
Primary dataset in original paper	KFall (multi-class)	SisFall binary + near-fall ADLs (D18/D19)
Fair baseline in our tables	—	<code>tier1_only</code> (same checkpoint/folds)

TABLE II
PER-FOLD COMPARISON OF TIER-1 (`TIER1_ONLY`) VS. FULL AGENTIC STACK (`GATE_KNN_LLM`) UNDER THE HOMOGENEOUS PROTOCOL ($n=1000$ STRATIFIED WINDOWS/FOLD WITH FORCED D18/D19). COST IS NORMALIZED PER 1000 WINDOWS.

Fold	n	Tier-1			gate_knn_llm			$\Delta F1$
		F1	Rec.	Cost/1k	F1	Rec.	Cost/1k	
0	1000	0.774	0.781	1042	0.892	0.986	156	+0.119
1	1000	0.737	0.748	1223	0.851	0.911	490	+0.114
2	1000	0.755	0.783	1066	0.874	0.998	134	+0.119
3	1000	0.766	0.732	1248	0.930	0.995	83	+0.164
4	1000	0.760	0.749	1175	0.887	0.993	136	+0.126

TABLE III
FIVE-FOLD MEAN $\pm$ STD ON SISFALL (HOMOGENEOUS $n=1000$). PAIRED DELTAS ARE `GATE_KNN_LLM`–`TIER1_ONLY`. WITH FIVE FOLDS, WILCOXON $p=0.0625$ IS AT THE DISCRETE MINIMUM FOR A COMPLETE SIGN AGREEMENT; WE THEREFORE EMPHASIZE BOOTSTRAP 95% CIs.

Mode	F1	Recall	Cost / 1000
<code>tier1_only</code>	0.759 $\pm$ 0.014	0.759 $\pm$ 0.022	1151 $\pm$ 93
<code>gate_knn_llm</code>	0.887$\pm$0.029	0.977$\pm$0.037	200$\pm$164
<i>Paired Δ (stack – Tier-1)</i>			
Metric	Mean Δ	Bootstrap 95% CI	Wilcoxon p
F1	+0.128	[0.117, 0.147]	0.0625
Recall	+0.218	[0.187, 0.248]	0.0625
Cost/1000	-951.000	[-1084.000, -824.800]	0.0625

TABLE IV
ABLATION LADDER ON FOLD 0 UNDER THE FULL EVALUATION BUDGET ($n=4000$). GATE OR k -NN ALONE DOES NOT YIELD THE HEADLINE GAIN; THE FULL STACK (`GATE_KNN_LLM`) IS REQUIRED.

Mode	F1	Recall	Cost	Near-fall FAR	Esc. F1
<code>tier1_only</code>	0.788	0.787	4596	0.219	0.000
<code>gate_only</code>	0.788	0.787	4596	0.219	0.175
<code>gate_knn</code>	0.789	0.764	4972	0.188	0.026
<code>gate_llm</code>	0.760	0.924	2491	0.625	0.445
<code>gate_knn_llm</code>	0.907	0.993	516	0.212	0.952

TABLE V
PAIRED LLM BACKEND COMPARISON ON SHARED AMBIGUOUS CASE IDS (UP TO 150 CASES/FOLD).

Fold	Heuristic F1	Mistral F1	Qwen2.5 F1
0	0.042	0.959	0.807
1	0.105	0.977	0.874
2	0.000	0.975	0.819
3	0.021	0.984	0.802
4	0.000	0.966	0.813
Mean	0.034	0.972	0.823

TABLE VI
TIER-1 BACKBONE ZOO ON FOLD 0 (DETECTOR-ONLY METRICS). PRIMARY PAPER BACKBONE IS `CNN_LSTM_ATT` (BHATTI-STYLE).

Model	F1	Recall	Spec.	Params
<code>cnn1d</code>	0.721	0.985	0.277	43,330
<code>lstm</code>	0.700	0.986	0.197	201,986
<code>cnn_lstm</code>	0.694	0.995	0.155	86,914
<code>cnn_lstm_attn</code>	0.710	0.996	0.216	564,930
<code>tcn</code>	0.727	0.988	0.295	89,282
<code>modern_tcn</code>	0.737	0.990	0.326	30,314
<code>tsmixer</code>	0.695	0.984	0.181	224,562
<code>patchst</code>	0.669	0.996	0.051	150,786
<code>transformer</code>	0.678	0.994	0.092	158,722

D. Backbone Context

Table VI reports detector-only Fold 0 metrics for the backbone zoo. We fix `cnn_lstm_attn` as the primary Tier-1 model for all agentic tables to keep comparisons fair to the Bhatti-style architecture.

VI. DISCUSSION

A. How We Differ from the Base Paper

Table I summarizes the conceptual gap: Bhatti *et al.* optimize multi-phase detection; we optimize a *decision system* around a detector. Fair numerical comparison in this manuscript is therefore `tier1_only` vs. `gate_knn_llm` under identical SisFall folds and checkpoints. The original Beyond Falls evaluation emphasizes KFall multi-class settings [4], [5]; SisFall is our primary runnable binary+near-fall corpus with D18/D19 instrumentation.

B. How We Outperform the Fair Baseline

Outperformance is threefold: (i) higher F1 and recall on every fold; (ii) large reduction in normalized response cost under FN-heavy weighting; (iii) high escalated-subset F1 when the gate defers. Near-fall FAR improvements are modest and are *not* claimed as the primary differentiator.

C. Limitations

Local LLMs add latency on escalated windows; we report escalation rates rather than claiming real-time free-living deployment. External validation on KFall remains future work pending raw data. With five folds, nonparametric *p*-values are coarse; bootstrap CIs are the primary uncertainty quantification. Detector-only F1 under our cost-sensitive operating point is intentionally not optimized as a leaderboard F1 ≥ 0.90 claim.

VII. CONCLUSION

We introduced a confidence-gated multi-agent wrapper for wearable fall detection that escalates ambiguous IMU windows to retrieval-augmented LLM reasoning and a cost-sensitive Action Agent. Under subject-independent five-fold SisFall evaluation with a locked protocol, the full stack consistently outperforms the same Bhatti-style Tier-1 detector in F1, recall, and normalized response cost, while ablations and paired LLM comparisons explain where the gains arise. This evidence supports a Q1-oriented narrative centered on *agentic safety layers around strong detectors*, complementary to backbone-centric fall detection research.

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APPENDIX A REPRODUCIBILITY NOTES

Primary artifacts: per-fold JSON under
 results/fold{0--4}/, aggregates
 results/aggregate_*, and
 briefs results/GUIDE_BRIEF.md, Pa-
 per tables can be regenerated via python
 scripts/generate_paper_tables.py.

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